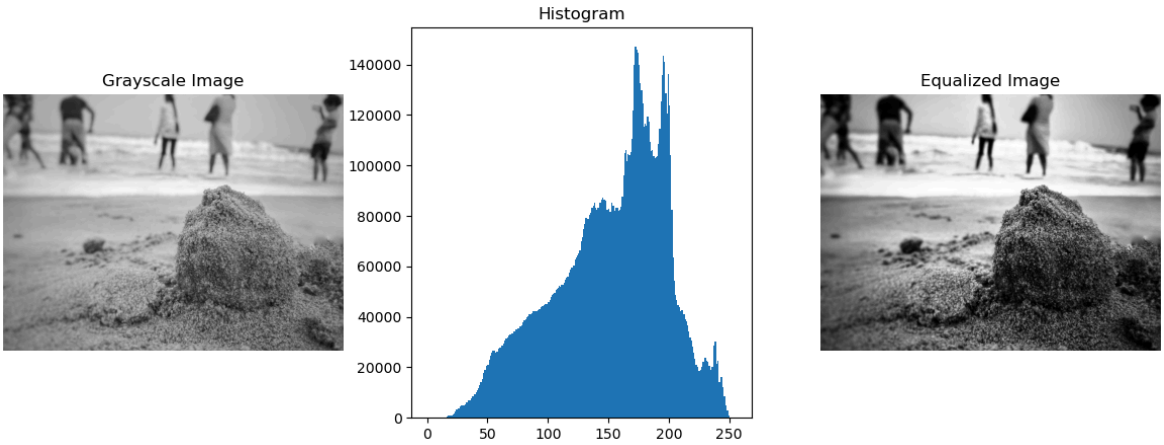
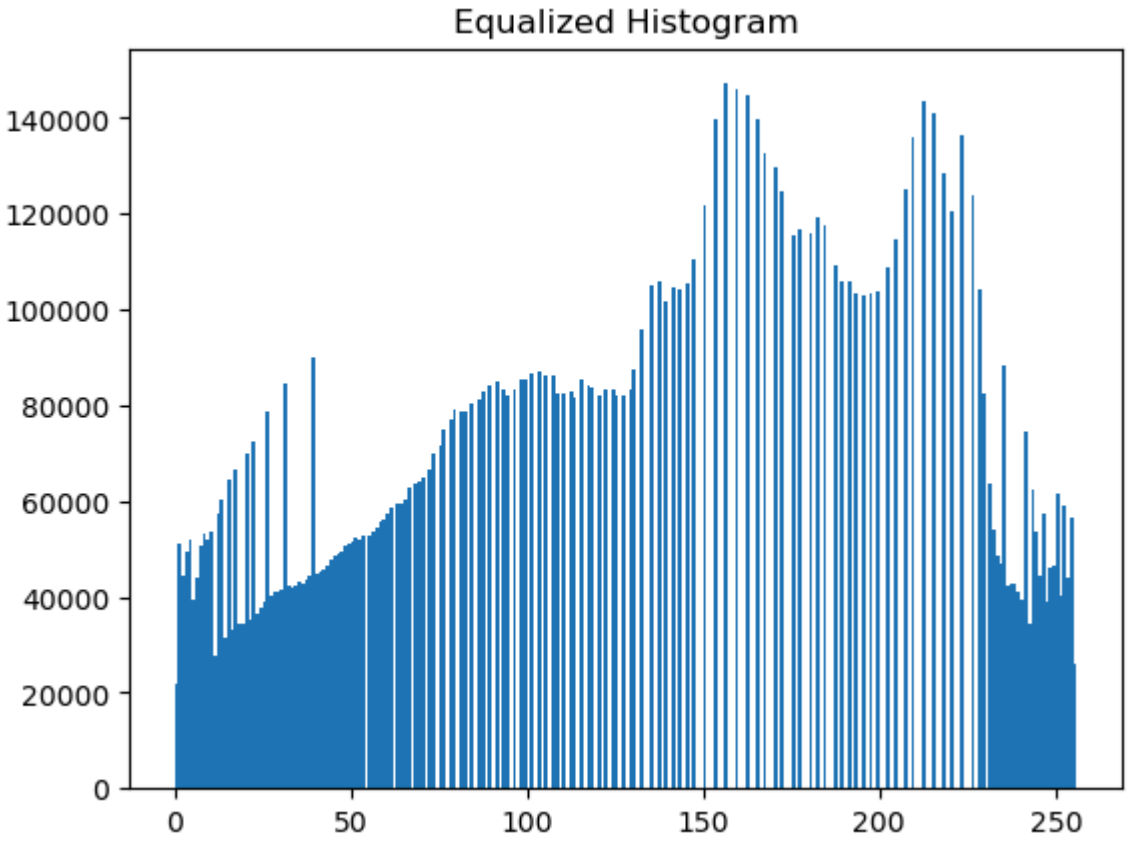
In [17]:
```
import cv2
import matplotlib.pyplot as plt

# Read image
img = cv2.imread("1.jpeg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# -------------------------------
# 1. Histogram
# -------------------------------
plt.figure(figsize=(20,5))

plt.subplot(1,4,1)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis("off")

plt.subplot(1,4,2)
plt.hist(gray.ravel(), 256, [0,256])
plt.title("Histogram")

# -------------------------------
# 2. Histogram Equalization
# -------------------------------
equalized = cv2.equalizeHist(gray)

plt.subplot(1,4,3)
plt.imshow(equalized, cmap='gray')
plt.title("Equalized Image")
plt.axis("off")

plt.show()

# Histogram of equalized image

plt.hist(equalized.ravel(), 256, [0,256])
plt.title("Equalized Histogram")
plt.show()
```




Equalized Histogram
140000
120000
100000
80000
60000
40000
20000
0
0
50
100
150
200
250